

Seabin Lee

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PROFESSIONAL SUMMARY

Robotics researcher specializing in large-scale **Multi-Robot Task Allocation (MRTA)** across multi-mobile-robot and multi-manipulator systems, with work on **Multi-Agent Path Finding (MAPF)** and **Human-Robot Interaction (HRI)** for scalable coordination and supervision. Project manager of funded research projects with ETRI, Hyundai Motor Group, and Doolim-Yaskawa.

EDUCATION

Sogang University <i>Ph.D. in Electronic Engineering (Advisor: Prof. Changjoo Nam)</i>	Seoul, South Korea Mar 2024 – Present
Sogang University <i>M.S. in Electronic Engineering (Advisor: Prof. Changjoo Nam)</i>	Seoul, South Korea Mar 2022 – Feb 2024
Sogang University <i>B.S. in Electronic Engineering</i>	Seoul, South Korea Mar 2016 – Feb 2022

PUBLICATIONS

Enhancing Situation Awareness in Multi-Robot Supervision: The Role of Adaptive Interfaces for Multiple Displays	2026
<i>Int. Journal of Human-Computer Interaction (IJHCI) S. Park, S. Lee, S. Park, C. Nam Published</i>	
Very Large-Scale Multi-Robot Task Allocation in Challenging Environments via Robot Redistribution	2025
<i>Robotics and Autonomous Systems (RAS) S. Lee, J. Sim, C. Nam Published Code</i>	
Coordination of Multiple Mobile Manipulators for Ordered Sorting of Cluttered Objects	2023
<i>IEEE/RSJ Int. Conf. on Intelligent Robots and Systems (IROS) J. Ahn, S. Lee, C. Nam Published</i>	

PREPRINTS & UNDER REVIEW

* Equal contribution

Real-Time Human-Adaptive Task Allocation for Multi-Human Multi-Robot Supervision	2026
<i>IEEE Trans. on Human-Machine Systems (T-HMS) S. Lee*, S. Park*, N. Kim, S. Park, H. Jung, C. Nam Under Review</i>	
HEART: Coordination of Heterogeneous Expert Agents for Physically Grounded Robotic Task Planning	2026
<i>IEEE Robotics and Automation Letters (RA-L) J. Lee*, S. Lee*, W. Lee, N. Kim, M. Kang, C. Nam Under Review arXiv</i>	
Reinforcement Learning for Robotic Insertion of Flexible Cables in Industrial Settings	2025
<i>IEEE Trans. on Automation Science and Engineering (T-ASE) J. Park, S. Lee, C. Park, W. Lee, C. Nam Under Review arXiv</i>	

PATENTS

Object Classification Method	Granted
<i>KR Patent No. 10-2917098 Registered Jan 20, 2026</i>	
Real-Time Human-Adaptive Task Allocation for Multi-Human Multi-Robot Supervision	Pending
<i>KR Application No. 10-2026-0048896 Filed Mar 18, 2026</i>	
Coordination of Heterogeneous Expert Agents for Physically Grounded Robotic Task Planning	Pending
<i>KR Application No. 10-2026-0048897 Filed Mar 18, 2026</i>	
Multi-Robot Task Allocation Method	Pending
<i>KR Application No. 10-2025-0140923 Filed Sep 29, 2025</i>	
Task Allocation Method, System and Program for Multi-Robot Considering Movement Path	Pending
<i>KR Application No. 10-2024-0100528 Filed Jul 29, 2024</i>	

RESEARCH PROJECTS

Development of a Continual Intelligence-enhancing RaaS Framework based on Human-Robot Shared Control	Jul 2024 – Present
<i>Government project, led by ETRI</i>	
<i>Project Manager</i>	
– Developed robot control task allocation algorithms considering human factors of remote operators – Constructed a simulation-based testbed using physics-based dynamics – Collected and analyzed human factor data through VR-integrated experiments	
Development of Harmonic-space Operation using Seamless Information Sharing among Large-scale Heterogeneous Vehicles	Jul 2024 – Present
<i>Government project, led by KETI</i>	
– Developing cooperative task allocation optimization for multiple vehicles considering mission and spatial information and vehicle-specific characteristics	

Multi-Manipulator Task Allocation for Defect Grinding

Apr 2025 – Sep 2025

*Industry project, Doolim-Yaskawa**Project Manager*

- Developed sanding task allocation algorithms while mitigating interference between multiple robot manipulators

General-Purpose Robotic Task and Motion Generation Software Platform

Oct 2025 – Present

Industry project, Samsung Electronics

- Building a multi-agent-based, general-purpose software platform that generates and manages task plans and motions for individual and joint tasks on heterogeneous robot hardware
- Developing task allocation algorithms for the platform's task planning, task execution, and autonomous learning processes

Development of Multi-Robot Task Allocation Algorithms for Efficient Robotic Services

Dec 2025 – Present

*Industry project, Hyundai Motor Group Robotics LAB**Project Manager*

- Developing task allocation algorithms for multi-robot delivery operations in multi-story buildings with elevator integration

Development of Autonomous Task Distribution and Physical Collaboration Framework for Multiple Heterogeneous Humanoid Robots

Apr 2026 – Present

Government project, led by KIST

- Developing mission planning and task distribution techniques for multiple humanoid robots

HONORS & AWARDS**Best Paper Award (Grand Prize), 2025 Sogang Electronic Engineering Paper Competition**

Nov 2025

Dept. of Electronic Engineering, Sogang University (BK21 FOUR) | Awarded for the RAS 2025 first-author paper

SKILLS & ADDITIONAL INFORMATION**Languages:** Python, C++, C# **Simulation & Robotics:** Isaac Sim, cuRobo, Unity, Blender **Tools:** Linux, Git**Additional:** Completed mandatory military service | No restrictions on overseas travel